

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Justification

LTZGLUE provides a GLUE-style suite of eight tasks covering binary and multi-class sentence- and token-level classification [Q11, Q67], which is a credible general-purpose NLU foundation for Luxembourgish. However, for the specific deployment — routing citizen correspondence to government departments — the taxonomies are misaligned. Topic classification covers only five RTL editorial categories (SPORTS, CULTURE, TECHNOLOGY, BUSINESS, ANIMALS) [Q48, Q153] with no administrative categories such as cross-border worker issues, housing, transport, tax, social security, or communal vs. national competency routing — all flagged as required in the elicitation and regional context. Intent detection labels are exclusively voice-assistant commands (alarm, reminder, weather) [Q154], with no administrative intents (complaint, document request, escalation). The benchmark contains no multi-label task formulation [Q67], whereas the deployment requires multi-label outputs [elicitation Q4]. The authors themselves acknowledge no unified benchmark currently exists for LTZ NLU [Q14]. Given IO is HIGH priority and the gaps span both omission (administrative categories) and irrelevance (voice-assistant intents) for this deployment, this constitutes significant taxonomic misalignment.

Strengths

- Provides eight diverse NLU task formulations including NER, topic classification, sentiment, RTE, and acceptability, establishing a baseline construct coverage for Luxembourgish [Q11, Q67]
- A substantial proportion of tasks are newly created for LTZ rather than direct translations, enabling some language-specific phenomena [Q69]
- NER schema includes PER, ORG, LOC, MISC, DATE [Q152] which has partial overlap with entities relevant to administrative correspondence (named persons, organizations, locations)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: The deployment requires categories for cross-border worker / frontaliere issues, housing/cost-of-living, transport, tax, social security, EU institutional matters, residency permits, communal services, public health, plus a state-vs-communal routing dimension and multi-label outputs with confidence scores [elicitation Q1, Q4; regional context administrative_domain_notes].

ID	Evidence
WEB-5	~231,290 frontaliere (~47% of employment) — citizen-input distribution will be heavily multilingual
WEB-2	OECD 2025: 50% of frontaliere from France, remainder split between Belgium and Germany — supports differentiated administrative-query profile

IO-2: Yes — none of LTZGLUE's topic classes (SPORTS, CULTURE, TECHNOLOGY, BUSINESS, ANIMALS) [Q48, Q153] correspond to administrative routing categories, and intent detection covers only voice-assistant commands [Q154]. No

state-vs-communal competency dimension or multi-label formulation exists [Q67].

IO-3: Yes — the voice-assistant intent labels (alarm, reminder, weather sub-intents) [Q154] are construct-irrelevant for administrative correspondence routing. The RTE logical-entailment framing [Q59] is unrelated to routing decisions, though it may have ancillary value for general LLM evaluation.

IO-4: Major content-validity gap: the benchmark cannot assess routing taxonomy performance for cross-border worker issues, housing, transport, tax, communal vs. national competencies — the categories that drive operational value [regional context benchmark_validity_risks_summary.input_ontology_gap]. The absence of multi-label formulation [Q67] additionally prevents evaluation of the deployment's required output structure.

Information Gaps

- Whether any LTZ-specific administrative-domain benchmark or taxonomy exists outside LTZGLUE — flagged as not searched in regional context
- Formal enumeration of communal vs. national competencies needed to define a routing taxonomy

Requires Expert Verification

- Final administrative topic taxonomy and multi-label structure must be co-designed with civil servants and legal/administrative experts

Remediation

Gap 1: No Luxembourg-administrative topic or intent taxonomy, no state-vs-communal routing axis, no multi-label formulation

Recommendation: Co-design an administrative topic taxonomy with national and communal civil servants covering at minimum: cross-border worker matters, housing/cost-of-living, transport, tax, social security, residency, communal services, EU institutional. Include a state-vs-communal routing dimension and a multi-label structure aligned with the deployment's actual case types.